

Project Analysis & Model Development – Summary Report

1. Project Objective

This project involves analyzing the dataset to uncover insights and building machine learning models to predict the target variable. The workflow covers EDA, preprocessing, model creation, and evaluation.

2. Dataset Overview

The dataset includes structured data with numerical and categorical features. Initial inspection revealed missing values, imbalance, and variations in feature distributions.

3. Exploratory Data Analysis (EDA)

- 1 Analyzed feature distributions and target variable behavior.
- 2 Identified important features and correlations.
- 3 Detected outliers and imbalance patterns.

4. Data Preprocessing

- 1 Handled missing and inconsistent data.
- 2 Encoded categorical variables and scaled numerical features.
- 3 Prepared clean training and testing datasets.

5. Model Creation

- 1 Developed baseline and advanced machine learning models.
- 2 Captured linear and non-linear relationships in data.
- 3 Compared multiple algorithms for performance.

6. Model Evaluation

Models were evaluated using accuracy, precision, recall, F1-score, and confusion matrices. The best-performing model was selected based on balanced performance.

7. Conclusion

The project highlights the importance of data understanding and systematic model development. Well-tuned models can provide reliable predictions and support informed decision-making.